Weather, Interconnections, and Energy Prices

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Project team
Author details to be completed

Abstract

Across 49 European market identities, we compare persistence, independent/pooled XGBoost, static/GRU MLP and self-loop/real GAT. Pooled XGBoost, static MLP and real GAT achieve 21.06, 21.19 and 20.97 EUR/MWh matched test MAE; the verified residual GRU achieves 20.98 in one seed. Graph gains are modest after controlling for pooled learning. A numerical audit invalidates the earlier negative GRU interpretation. These are exploratory retrospective benchmarks with unresolved source vintages.

1 Approach

At 12:00 Europe/Brussels on calendar day $D-1$, predict every hourly day-ahead price of delivery day D, including 23/25-hour daylight-saving days. Weekly persistence copies the same market's price 168 elapsed hours earlier. Independent and pooled XGBoost [1] distinguish shared learning from cross-market graph messages. A pooled MLP provides a classical neural reference. Self-loop GATv2 [2] retains the architecture without cross-market messages; real GAT adds physical pairs observed during training. These GATs share features and training settings. Node-lifetime masks preserve historical geography. JAO constraints enter as flat summaries, not extra graph nodes.

2 Experiments

Data and partitions. The canonical master contains 5,887,954 zone-hours over 62 historical price-area identities, January 2016–October 2026. ENTSO-E prices/load/flows use pinned v1.2 releases; other sources are an ERA5-lineage weather export and original CWE/CORE JAO observations. Weather's requested v1.1 release identity is unverified. Missing values, negative prices and separate DE_AT_LU/DE_LU identities are preserved; 49 identities have labels in all required partitions. Targets are never filled or filtered for feature completeness. Fit uses January 2016–March 2024, validation April 2024–March 2025, and test April 2025–September 2026 by Brussels delivery date. October stays unassigned.

Static controls. All five static learned controls use identical richer inputs: local hour/weekday/month, guarded 24/168-hour prices, load forecast, four lag-48 weather variables and three lag-48 JAO summaries (constraint count, minimum RAM ratio, domain presence). Trees retain NaNs; neural imputation/scaling fits training features only and includes missingness indicators. Models fit $z = \operatorname{asinh}(p/50)$ with L1 loss and invert with $50\sinh(\hat{z})$. Trees use depth six, at most 600 rounds and 50-round validation patience. MLP has two ELU hidden layers matched by parameter count to GAT; GAT has two layers, 64 channels and four heads. Both use Adam 0.001, dropout 0.1, at most 30 epochs and patience five. All five controls use seeds 224–228.

Partner-inspired temporal model. The GRU–MLP adapts Sasha's recipe on our canonical targets, rather than importing his separate 43-zone score. A 64-unit GRU encodes 168 completed hours of price lags, load, derived net inflow and four weather channels. Two 128-unit ReLU MLP layers combine its state with 11 day inputs. The target is $r = z - \operatorname{asinh}(b/50)$, with weekly-price base b, else guarded daily price, else zero; this fallback reconstructs predictions only. Residual L1 is augmented by 0.5 times connected-zone residual-contrast L1. Day weather uses lag 48. Seed 0,

16-day batches, Adam 0.001 and patience eight select epoch 29 of 37. This is a different recipe and one seed, not a controlled GRU or weather ablation.

Metrics. MAE/RMSE are EUR/MWh after inversion. Relative MAE divides model MAE by persistence MAE on identical hours: 0.72 means 28% less error, not 28% of the price. Table 1 uses 428,213 validation and 633,885 test labels with weekly history. Full scoring retains all 428,427 validation and 634,578 test labels, including unavailable histories. Validation selects checkpoints; no best test seed is chosen.

Forecasting method	Seeds	Val. MAE	Test MAE	Test rMAE
Weekly persistence	–	30.18	29.13	1.0000
Per-zone XGBoost	5	21.23 ± 0.05	22.11 ± 0.03	0.7589
Pooled XGBoost	5	20.74 ± 0.01	21.06 ± 0.01	0.7228
Static pooled MLP	5	21.44 ± 0.23	21.19 ± 0.21	0.7272
GRU + MLP (partner)†	1	**20.32**	20.98	0.7202
Self-loop GAT	5	21.27 ± 0.09	21.06 ± 0.08	0.7230
Real-graph GAT	5	20.98 ± 0.11	$\mathbf{20.97 \pm 0.13}$	**0.7199**

Table 1: Same 49-market cohort and matched hours. Static learned rows: five-seed mean $\pm$ sample SD, not ensembles. †: one verified seed, different inputs/target/loss. Bold denotes the smallest reported mean, not significance or a controlled GRU advantage. Lower is better.

Results. Pooling reduces XGBoost test MAE from 22.11 to 21.06. Real GAT improves test MAE by 0.21 versus static MLP and 0.09 versus self-loop GAT. Corresponding validation-week improvement intervals are [0.20, 0.86] and [0.13, 0.46] EUR/MWh (2,000 shared-week resamples, conditional on fitted models). Pooled XGBoost has better validation MAE than real GAT; their small test gap does not justify retrospective model choice. The GRU has similar matched MAE but all-target test RMSE 47.02 versus 34.92 for real GAT. Gains vary by market; graph superiority is not universal.

Numerical audit. The original controlled GRU–MLP/GAT runs are omitted: all ten checkpoints disagree with CPU on sampled shuffled training-shaped inputs under fused ROCm, despite replaying saved consecutive inference batches. Native GPU agrees with CPU; the negative temporal conclusion is withdrawn. The accepted partner-inspired run uses native GPU operations and passes sampled CPU float64, checkpoint-replay, target-signature and saved-score checks. This is not an exhaustive backend proof.

Availability and limitations. ERA5 is retrospective reanalysis with roughly five-day release latency [3], incompatible with recent lag-48 inputs. Load/JAO/price revision vintages remain unresolved. Exact timestamp joins and autumn-DST guards prevent lag24 from reaching the target delivery day, but cannot reconstruct publication records. Scores are conditional, not certified operational forecasts. Test data were previously explored; intervals are not confirmatory evidence.

3 Future work

Retrain our original temporal controls with native GPU operations; repeat the residual GRU across seeds and ablate its residual target and contrast loss. Build a daily service that records inputs, versions and predictions before Brussels noon, then scores auction outcomes. Start with persistence and price-only pooled XGBoost; monitor rolling rMAE, coverage, deadline failures and fallbacks. Add timely weather observations/archived forecasts and graph challengers after establishing this prospective baseline.

References

[1] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. *KDD*, 2016. arXiv:1603.02754.

[2] S. Brody et al. How attentive are graph attention networks? *ICLR*, 2022. arXiv:2105.14491.

[3] ECMWF. ERA5 hourly time-series product description. Accessed November 9, 2026.